Mohneet Sandhu

+1 (240) 438-0963 mohneet@umd.edu MohneetSandhu MohneetKaur mohneetkaur.com

Education

University of Maryland, College Park, MD **Aug 2024 - May 2026**
Masters in Data Science
Relevant courses: DSA, NLP, Machine Learning, Cloud Computing, LLM.

Narsee Monjee Institute of Management Studies, Maharashtra, India **Jun 2020 - Jul 2023**
Bachelor of Science in Applied Statistics & Analytics: 3.91/4 (Dean's List)
Relevant courses: Python, R, Experimental Design, SQL, Hypothesis Testing, Bayesian Statistics, Linear Algebra.

Skills

Programming Languages: Python (PyTorch, TensorFlow), R, SAS, JMP, Java, C++, SPSS, SQL (PL/SQL, PostgreSQL, NoSQL), MATLAB.
Data Science & Analytics: A/BTesting, Experimental Design, Statistical Modeling,Tableau, Power BI, Alteryx, MS Office.
Machine Learning: CNN, RNN, GAN, NLP, GNN, VAE, RAG, Transformers, Time Series, Computer Vision, Reinforcement Learning.
Big Data Technologies: Hadoop, Spark, Kafka, Dataiku, BigQuery, Hive, ETL.
Cloud Technologies: AWS, Azure, Google Cloud Platform, Salesforce, Redshift, Snowflake, GitHub, Docker, Databricks, Airflow.
Generative AI Tools: Open AI, Langchain, LangGraph, LlamaIndex, Gemini, Hugging Face.

Experience

Amazon: Data Scientist Intern — May 2025 - Aug 2025

- Automated LLM-driven generation of station-level reports identifying root causes of DEA misses, reducing manual investigation **~85%**.
- Processed 100M+ events/day using cloud-native AWS, SQL, and event-driven distributed pipelines for low-latency diagnostics.
- Developed production LLM workflows with prompt engineering, RAG, embeddings, and agent orchestration aligned with SDLC practices.
- Optimized agentic LLM analytics on Redshift using DSPy, reducing misattribution ~28% and false positives ~22% in risk diagnostics.

University of Maryland, College Park: Data Scientist — Jan 2025 (Present)

- Built scalable LLM pipelines using Python, SQL,Java, Databricks, AWS & Azure cloud services supporting **10K+** production analytics workflows.
- Developed AI chatbots and agent workflows using Prompt Engineering, LangChain, LLM orchestration, & vector retrieval for context-aware search.
- Applied statistical modeling using distribution analysis, hypothesis testing, regression, and confidence interval inference to drive insights.
- Implemented agent architectures using MCP orchestration and MLOps practices across enterprise systems.
- Developed distributed ML pipelines using Spark, Hadoop, Kafka streaming, and SQL data stores for batch and real-time processing systems.
- Applied GenAI methods including LLM fine-tuning, embeddings, RAG workflows, HuggingFace/PyTorch, and retrieval-based automation.
- Built dashboards using Tableau, Power BI, Excel and Python to visualize model insights and KPIs, enabling faster data-driven decisions.
- Built ETL pipelines using BigQuery, Airflow, dbt, Kafka, and Docker and Kubernetes, reducing data latency by **~50%**.

Innomatics Research Labs: Data Science Intern — Feb 2023 – May 2023

- Fine-tuned RoBERTa (PyTorch) on 500K+ SaaS/B2B financial logs, boosting fraud detection precision by **30%**, with LangSmith tracing.
- Built real-time NLP pipelines with Hugging Face and Python, reducing false positives by **37%** in risk flagging.
- Delivered Power BI dashboards and SQL validation reports, accelerating team risk response time by **2x**.

Technocolabs Software: Data Analyst Intern — Jul 2022 – Aug 2022

- Designed containerized ETL pipelines using SQL, Python, Spark, and Docker, processing **100K+** records daily for robotic telemetry analysis.
- Developed real-time dashboards using Power BI, Redshift, and ggplot2 to monitor operational and scheduling KPIs.
- Used Hadoop and Databricks for distributed computation and scalable workflows, reducing processing time by **3×**.

Suvidha Foundation: Machine Learning Intern — May 2022 – Jun 2022

- Built an LLM agent with Langchain to automate security policy reviews, detecting inconsistencies and outdated configurations.
- Developed GPT-2 prompt pipelines to extract regulatory signals from Snowflake data using NLP-driven semantic parsing & risk evaluation.
- Reduced manual audit effort by **40%** by building Python-based validation engines with rule orchestration and automated evaluation pipelines.

Projects

LLM-Powered Hypothesis Generator | Google Cloud Multi-Agent System

- Built a Gemini-1.5 multi-agent pipeline using ADK, LangGraph, and Vertex AI to generate structured hypotheses from multimodal data.
- Integrated RAG pipelines with BigQuery + GCS embeddings, vector search (FAISS), semantic clustering (HDBSCAN), & multi-modal evidence fusion, enabling production of 100–150 auto-ranked hypotheses per run with linked citations and signal attribution.
- Architected DSPy optimization & multi-agent orchestration pipelines, improving efficiency & reducing hypothesis generation time by **~60%**.

AI Clinical Guidance & Triage System | Multi-Agent Generative AI, RAG, MCPs

- Built a multi-agent reasoning system using OpenAI MCPs, tool-calling, & domain-tuned LLMs for symptom triage & differential diagnosis.
- Integrated Integrated biomedical RAG (BioBERT, FAISS, PubMed/CDC/WHO MCPs, SNOMED), boosting guideline adherence by **33%**.